

Ticket System v3

Complete manual — project rundown & user guide

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1) Project rundown (technical)

1.1 Purpose

Ticket System v3 is a full-stack **service desk / ticketing** application for submitting requests, assigning work to teams and personnel, tracking **SLA** targets, escalating breached items, managing **KPI** maintenance tasks, and collecting **customer verification and feedback** after resolution.

1.2 Technology stack

Layer	Technology
Application	Next.js (App Router), React , Tailwind CSS
API	Next.js Route Handlers (src/app/api/**)
Database	PostgreSQL with Prisma ORM
Authentication	NextAuth.js — local credentials and optional Google OAuth
Email	SMTP (e.g. Brevo) for resolution and verification flows

1.3 Repository layout (high level)

- **src/app/** — Pages and API routes (App Router).
- **src/components/** — Shared UI (navigation, portal shell, dashboards).
- **src/lib/** — Business logic: auth, SLA, email, ticket actions, portal accounts.
- **prisma/** — Schema, migrations, seed.
- **ecosystem.config.cjs** — **PM2** process definition for production on Node hosts.

1.4 Core concepts

- **Ticket** — Single customer issue with status (Open → In progress → Resolved → Closed, plus Pending info / Escalated as needed).
 - **Priority** — Includes a “**Set Priority Level**” (**UNSET**) default until staff sets **Low / Medium / High / Urgent**; work typically cannot move to **In progress** until priority is set (policy enforced in API/board).
 - **SLA** — First-response and resolution due times derive from **SlaPolicy** per priority; automated sweep can escalate breached unresolved tickets (Admin-triggered API).
 - **Teams & agents** — Tickets route to teams; **Personnel** work assigned queues from **/agent**.
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2) Architecture & data

2.1 Request flow (simplified)

1. User authenticates → session carries **role** and **email**.
2. Customer submits ticket → **POST /api/tickets** creates row, SLA timestamps, activity log.
3. Staff manage lifecycle → **PATCH /api/tickets/[id]** (status, priority, escalation, resolution).
4. On **Resolved**, system may email the **requestor inbox** (SMTP); customer **verifies** and submits **rating** where configured.

2.2 Notification email routing

- **Google sign-in:** notification and verification emails target the **Google account email** (authProvider: google).
- **Portal / credentials:** requestorEmail on the ticket aligns with the **portal account work email** stored in the database (with safeguards); synthetic portal identities may require an explicit notification email on intake.

2.3 Pending resolution confirmation (customers)

If a customer has a **resolved** ticket that still needs **verification and mandatory rating** (no feedback record yet), **new ticket creation may be blocked** until that flow is completed. The UI surfaces a **notification** and links to the **verification** path (e.g. /tickets/[id]/verification).

3) Authentication & roles

3.1 Sign-in

- **URL:** /signin
- Methods: **local credentials** (username or email + password) and **Google** (if configured).

3.2 Role resolution (typical)

1. Portal account role from the database when an account exists.
2. Env lists **ADMIN_EMAILS** / **AGENT_EMAILS** for bootstrap mapping.
3. Default **Customer** where no other rule applies.

3.3 Role summary

Role	Typical access
Customer	Dashboard (/), Active Tickets , Create ticket , knowledge base, verification/rating, /admin/account.
Personnel	/my-requests, Ticket and Task boards (/agent), insights, optional assignment board.
Admin / SuperAdmin	Ticket dashboard, create requests, personnel, all board tabs, priority alerts, task definitions, insights, SLA sweep.

Head / privileged portal flags may grant coordination powers (e.g. escalation) per implementation.

4) Deployment & operations

4.1 Environment

Copy `.env.example` to `.env`. Key groups: `DATABASE_URL`, **NextAuth** secret and URL, optional **Google** client settings, **SMTP (Brevo)** for mail, `APP_BASE_URL` / `NEXTAUTH_URL` for correct email links, optional admin email lists.

4.2 Database

- **Prisma:** `npx prisma migrate deploy` or `npx prisma db push` (depending on how the environment was baselined).
- **Helpdesk Google Form CSV (Insights task metrics):** after deploy, upload the sheet export once per environment — `npm run db:apply-helpdesk-csv path/to/"IT SALF - HELPDESK.csv"` (spreadsheet **Completed** is counted as **For confirmation** alongside **Closed**).
- **Seed (optional):** `npm run db:seed` for demo data.
- One-off: `npx tsx scripts/ensure-unset-priority-data.ts` if using **UNSET** priority and SLA rows need ensuring after `db push`.

4.3 Production (Node + PM2)

- Build: `npm run build`
- Start: `pm2 start ecosystem.config.cjs` (or `npm run pm2:start`)
- Reload after deploy: `npm run deploy:pm2` or `pm2 restart ticket_system_v3 --update-env`

- See repository `DEPLOY_NGINX.md`, `DEPLOY_NGINX_WINDOWS.md`, or `DEPLOY_CPANEL.md` for host-specific steps.
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5) Customer user guide

5.1 First steps

1. Open `/signin` (legacy `/customer/signin` redirects here; `/signup` when self-registration is enabled).
2. Use the account your organization provided, or register per policy.
3. After sign-in, the **Dashboard** (`/`) shows a kanban of your tickets.

Portal navigation

Area	Path
Dashboard	<code>/</code>
Active Tickets	<code>/my-tickets</code>
Knowledge Base	<code>/tickets/knowledge</code>
Create Request	<code>/tickets/new</code>

Use the **notification bell** for updates and verification reminders.

5.2 Submit a ticket

1. Go to `/tickets/new`.
2. Complete **department / business unit**, **name**, and **issue** description.
3. Attach screenshots if helpful (image limits apply).
4. If you use **Google**, emails for resolution/verification go to your **Google address**. If you use a **portal** account, notification email follows your **registered work email** (optional override only when it matches your portal email).
5. Submit; track from `/my-tickets` or the dashboard.

5.3 Track and participate

- Open a ticket from **Active Tickets**, the dashboard kanban, or a shared link.
- Read the **activity timeline** and **conversation** thread.
- Dashboard columns group work as **Open**, **In progress**, and **For confirmation** (resolved, awaiting your verify/rate).
- If **new requests are blocked** because a **resolved** ticket awaits your **confirmation and rating**, open the **notification** or the linked **verification** page and complete the steps.

5.4 Resolution, verification, and rating

1. When staff marks the ticket **Resolved**, you may receive email with verify / reject links.

2. In the portal, complete **verification** and **star rating / feedback** when prompted.
3. Ratings of **3 stars or below** require written feedback before closure.
4. Until mandatory steps are done, **creating another ticket** may be disabled **policy**.

5.5 Ticket statuses (customer view)

Status	Meaning (plain language)
Open	Received; queued / triaged.
In progress	Actively worked.
Pending info	Waiting on information (implementation may vary).
Escalated	Elevated for leadership or specialized handling.
Resolved	Solution delivered; you should verify and rate.
Closed	Closed after confirmation / feedback per policy.

5.6 My account

Open `/admin/account` → **Security** tab (customers, personnel, and admins). **Security** changes (username, email, password) sign you out and require a fresh sign-in. Submit suspension/deletion requests for admin review when needed.

6) Staff & personnel guide

6.1 Daily routine

1. Sign in at `/signin`.
2. **Personnel** land on `/agent`; personal ticket list is `/my-requests`.
3. Use board tabs: **Ticket Board** (`/agent?board=ticket`) and **Task Board** (`/agent?board=kpi`).
4. Review `/insights` — ticket metrics, task metrics, and (coordinators only) **My Task Management**.

6.2 Working a ticket

1. Confirm assignment and priority (**set priority** if still **UNSET** — otherwise board/API may block **In progress**).
2. Update status honestly (drag-and-drop where enabled).
3. Use **request more information** per policy (may log activity **without** forcing “pending info” status depending on configuration).
4. Resolve with clear **resolution notes**; customer receives email to **verify**.
5. Staff ticket workspace: `/agent/tickets/[id]`.

6.3 KPI / task hygiene

- Complete checklist items where you are the KPI card assignee, or where an individual sub-KPI is assigned directly to you.
- Admins/coordinators can assign individual sub-KPIs to other personnel from inside each Task Board card.
- Non-IT Project sub-KPIs can store one **before** screenshot and one **after** screenshot for evidence; uploads must be **JPEG or PNG** and no larger than **10MB** each.
- Respect recurrence boundaries (daily / weekly / monthly cycles); period boundaries use browser/reporting timezone in API calls.
- **Recurring tasks:** flat or **segmented** sub-KPI checklists; checklists reset on period rollover.
- **IT Project Implementation:** non-recurring; phases and per-sub-task due dates (**MM/DD/YYYY** in UI); record **actual date** when completing sub-tasks. **Delayed** column applies only to this pillar (late sub-task or actual after due); fully complete but late work stays in **Delayed**, not **Done**.
- Other recurring KPIs: late completion may show **Done** with a delayed indicator.

6.4 Metrics & Reports (/insights)

Tab	Audience	Content
Ticket metrics	Admin, Personnel	SLA, volume, CSAT, charts (date range)
Task metrics	Admin, Personnel	Helpdesk / checklist pillar metrics
Task Management	Personnel coordinators	KPI definitions (when canAccessAssignmentBoard)

Admins define tasks on **Task Board** (/agent?board=kpi), not the Insights Task Management tab.

7) Admin & SuperAdmin guide

7.1 Oversight

- / — Ticket dashboard (on-duty, activity, priority stack).
- /admin/ticket-requests — Create requests on behalf of users.
- /admin/personnel — Personnel registry and assignment readiness.
- /admin/manual-assignment — Assignment board (drag to personnel lanes).
- /agent?board=company — Company board.
- /agent?board=ticket — Ticket board.
- /agent?board=kpi — Task board + **Task Definition** console.
- /admin/escalation-triggers — **Priority alerts** (escalation configuration).
- /insights — Metrics & reports.
- /process — Process / lifecycle reference.
- /reports — Reporting views as implemented.

7.2 SLA sweep

- **POST /api/sla/sweep** (secured; Admin context) — scans for SLA breaches and escalates/logs per policy.

7.3 Accounts

- **/admin/account-management** — Portal account administration where deployed.

8) Account security (My Account)

All roles with portal access use **/admin/account** (Security tab).

Change username

1. Security tab → new username, current password → **Update username**.
2. You are signed out; sign in with the new username.
3. Rules: length and character constraints shown in UI.

Change email / password

Similar pattern: confirm current password; after success, **sign in again** with new credentials.

Account requests

Suspension or deletion requests may be submitted for admin review.

9) Troubleshooting

Symptom	What to try
Cannot sign in	Verify credentials, caps lock, Google sign-in status if used, and account active state.
“Forbidden” or missing menu	Your role may not include that feature — contact Admin.
Cannot create new ticket	Complete pending resolved verification/rating on existing ticket.
Email link expired / invalid	Request staff to re-send resolution flow or open ticket in portal.
Cannot move ticket to In progress	Set priority above Set Priority Level (UNSET) .
Task stuck in Delayed (IT Project)	Complete overdue sub-tasks or correct actual/due dates.
Cannot edit KPI checklist	Confirm you are the assignee; IT Project requires per-sub-task actual dates.
Database / deploy errors	Check DATABASE_URL , run Prisma migrate or push, rebuild, restart PM2.

10) Quick reference — URLs

Path	Description
/signin	Sign in
/signup	Self-registration (if enabled)
/	Customer dashboard or admin ticket dashboard
/my-tickets	Customer active tickets
/my-requests	Personnel ticket dashboard
/tickets/new	New ticket
/tickets/knowledge	Knowledge base
/tickets/[id]	Ticket detail
/tickets/[id]/verification	Resolution verification
/tickets/[id]/rate	Rating / feedback
/agent	Staff boards (default ticket view)
/agent?board=ticket	Ticket board
/agent?board=kpi	Task board
/agent?board=company	Company board (admin)
/agent/tickets/[id]	Staff ticket workspace
/admin/manual-assignment	Assignment board
/admin/ticket-requests	Admin create requests
/insights	Metrics & reports
/process	Process / lifecycle info
/reports	Reports
/admin/personnel	Personnel admin
/admin/escalation-triggers	Priority alerts
/admin/account-management	Portal account admin
/admin/account	My Account & security

End of document.